

Class 10 · Practice Test (Maths and Science)

Time: 1 hour · Maximum marks: 26 · Answer all questions. Show your working.

Q1.	The middle term of the A.P.: 10, 7, 4, ..., -62 is: (A) -26 (B) -29 (C) -32 (D) -35	[1]
Q2.	Assertion (A): In an ecosystem, 1% of energy is available for transfer from one trophic level to the next. Reason (R): An average of 10% of the food eaten by organisms is available for flow in the form of chemical energy. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, and R is not the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true.	[1]
Q3.	Find the area of a sector of a circle with diameter 28 cm, if the length of the corresponding arc is 22 cm.	[2]
Q4.	Heat produced per second due to a current in a resistor of $4\ \Omega$ is 400 joules. Calculate the potential difference across the resistor.	[2]
Q5.	In what ratio does the x-axis divide the line segment joining the points (-4, -6) and (-1, 7)? Also, find the coordinates of the point of division.	[3]
Q6.	The lowest part of the ear, called earlobe, is closely attached to the side of the head in some of us (attached earlobe), and not in others (free earlobe). Attached and free earlobes are two variants found in human populations. The gene for free earlobe is dominant over attached earlobes. (a) A man with attached earlobes marries a woman having free earlobes. 50% of their children have free earlobes and 50% have attached earlobes. Explain the inheritance of this trait and write the trait combinations of the progeny. [2] (b) Write the gene combinations of the father and the mother in the above case. [1]	[3]
Q7.	A hormone 'X' is secreted in blood when a person is under scary situation. (a) Identify the hormone 'X' and the gland that secretes it. [1] (b) Explain its role in dealing with scary or emergency situations. [2]	[3]
Q8.	Tara scored 40 marks in a test, getting 3 marks for each right answer and losing 1 mark for each wrong answer. Had 4 marks been awarded for each correct answer and 2 marks been deducted for each wrong answer, then Tara would have scored 50 marks. Assuming that Tara attempted all questions, find the total number of questions in the test.	[5]

- Q9. Case study: A school conducted a weekly test for Class X students, before the commencement of the pre-board examination and recorded their scores (out of 50). To analyse performance patterns, the academic coordinator grouped the marks into intervals. The grouped frequency distribution is as below: [4]

Marks obtained	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
Number of students	3	6	12	15	14

Based on the above information, answer the following questions:

- (i) Identify the class with the most common performance range. [1]
- (ii) Find the class interval containing the median. [1]
- (iii) What is the average performance of the students? [2]

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- Q10. A. Due to narrowing of the bile duct, the bile secretion from the liver to the small intestine was obstructed. How will the process of digestion be affected? [1] [2]
- B. How does the length of the intestine depend on the nature of food habit in different animals? [1]
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